**List of JsonObjects output by the Plan Generator**

(27 August, 2015 Shirley Crompton, STFC)

Note: The table applies to update and create operations. If an object is being removed, only the name attribute will be provided.

| Object | Attribute | Type | Optional | Description |
|---|---|---|---|---|
| ApplicationTask | name | String | | Application name as in Camel model, e.g. Scalarm |
| | objType | String | | hardcoded to "application" |
| | description | String | yes | Description of the application. |
| | version | String | | Application version. |
| | owner | String | | Organisation that owns this application. |
| ApplicationInstanceTask | name | String | | Application name as in Camel model appended with Instance, e.g. ScalarmInstance |
| | objType | String | | hardcoded to "applicationInstance" |
| | type | String | | Application type name, i.e. the Application name |
| VMInstanceTask | name | String | | VM Instance name |
| | objType | String | | hardcoded to "vmInstance" |
| | type | String | | VM/Component type name |
| | cloud | String | | Cloud name |
| | VMTypeCloudProviderId | String | | VM type cloud provide Id |
| | locations | String Array | yes | Cloud provider location. |
| | driver | String | | Driver class for a particular cloud provider. Note that the value may be an empty String. |
| | endpoint | String | | Endpoint for the cloud provider |
| | credential | Json object | | This currently contains two key-value pairs : username and password Strings |
| | defaultCredential | Json object | | Contains key-value pairs: defaultLoginName and defaultLoginPassword Strings |
| | ~~cloudLocation~~ | ~~String~~ | ~~yes~~ | ~~Geographical region of the CloudLocation. It may not be populated. ()A dataCentre's location is a choice of cloudLocation, geographicalRegion, country.)~~ |
| | vmType | String | | VM flavour name |
| | VMImageId | String | | VM image id |
| | VMOS | String | yes | VM OS image name |
| | OSVendorType | String | | VM Provider type String |
| | OSArchitecture | String | | VM OS architecture |
| | VMMemory | int | yes | VM image capacity |
| | VMStorage | int | yes | VM storage capacity |
| | VMCores | int | yes | VM core number |
| | providedHostInstances | String Array | | Array of provided host instance names |
| | providedCommunicationInstances | String Array | yes | Array of provided communication instance names |
| VMTypeTask | name | String | | VM Type name |
| | objType | String | | hardcoded to "VM" |
| | osImage | String | yes | OS image name. An object only contains either osImage or os. Not both. |
| | os | String | yes | OS name. An object only contains either osImage or os. Not both. |
| | ~~os64bit~~ | ~~boolean~~ | ~~yes~~ | ~~True if IS is 64bit, else false. This is an attribute of os. (removed 23Nov15, obsolete, use the OSArchitecture in VMInstanceTask)~~ |
| | configName | String | yes | (Configuration) name |
| | downloadCmd | String | yes | (Configuration) download command |
| | configureCmd | String | yes | (Configuration) configure command |
| | installCmd | String | yes | (Configuration) install command |
| | startCmd | String | yes | (Configuration) start command |
| | StopCmd | String | yes | (Configuration) stop command |
| | uploadCmd | String | yes | (Configuration) upload command |
| | providedHosts | String Array | | Array of provided host names |
| | providedCommunications | String Array | yes | Array of provided communication names |
| ComponentInstanceTask | name | String | | Internal component instance name |
| | objType | String | | hardcoded to "internalComponentInstance" |
| | type | String | | Internal component type name |
| | ~~initialInstance~~ | ~~int~~ | | ~~Initial instance. Currently hardcoded to 1. This attribute may be removed in a later release.~~ |
| | providedHostInstances | String Array | yes | Array of provided host instance names |
| | requiredHostInstance | String | | Required communication instance name |
| | providedCommunicationInstances | String Array | yes | Array of provided communication instance names |
| | requiredCommunicationInstances | String Array | | Array of required communication instance names |
| ComponentTypeTask | name | String | | Internal component type name |
| | objType | String | | hardcoded to "internalComponent" |
| | configName | String | yes | (Configuration) name |
| | downloadCmd | String | yes | (Configuration) download command |
| | configureCmd | String | yes | (Configuration) configure command |
| | installCmd | String | yes | (Configuration) install command |
| | startCmd | String | yes | (Configuration) start command |
| | StopCmd | String | yes | (Configuration) stop command |
| | uploadCmd | String | yes | (Configuration) upload command |
| | providedHosts | String Array | yes | Array of provided host names |

| | | | | |
|---|---|---|---|---|
| | requiredHost | String | | Required host name |
| | providedCommunications | String Array | yes | Array of provided host names |
| | requiredCommunications | String Array | | Array of required communication  names |
| CommunicationInstanceTask | name | String | | Communication instance name |
| | objType | String | | hardcoded to "communicationInstance" |
| | type | String | | Communication type name |
| | providerInstance | String | | Communication provider instance name |
| | providerCompInstanceTask | String | yes | ComponentInstanceTask name - owner of the providerInstance |
| | consumerInstance | String | | Communication consumer instance name |
| | consumerCompInstanceTask | String | yes | ComponentInstanceTask name - owner of the consumerInstance |
| CommunicationTypeTask | name | String | | Communication type name |
| | objType | String | | hardcoded to "communication" |
| | communicationType | String | | LOCAL, REMOTE, ANY |
| | provider | String | | Provider name |
| | providerCompTypeTask | String | yes | ComponentTypeTask name - owner of the Provider |
| | providerPort | int | | Provider port number |
| | providedPortconfigName | String | yes | (Configuration) name |
| | providedPortdownloadCmd | String | yes | (Configuration) download command |
| | providedPortconfigureCmd | String | yes | (Configuration) configure command |
| | providedPortinstallCmd | String | yes | (Configuration) install command |
| | providedPortstartCmd | String | yes | (Configuration) start command |
| | providedPortStopCmd | String | yes | (Configuration) stop command |
| | providedPortuploadCmd | String | yes | (Configuration) upload command |
| | consumer | String | | Consumer name |
| | consumerCompTypeTask | String | yes | ComponntTypeTask name - owner of the Consumer |
| | isMandatory | boolean | | True if consumer depends on this communication, else false |
| | consumerPort | int | | Consumer port number |
| | requiredPortconfigName | String | yes | (Configuration) name |
| | requiredPortdownloadCmd | String | yes | (Configuration) download command |
| | requiredPortconfigureCmd | String | yes | (Configuration) configure command |
| | requiredPortinstallCmd | String | yes | (Configuration) install command |
| | requiredPortstartCmd | String | yes | (Configuration) start command |
| | requiredPortStopCmd | String | yes | (Configuration) stop command |
| | requiredPortuploadCmd | String | yes | (Configuration) upload command |
| HostingInstanceTask | name | String | | Hosting instance name |
| | objType | String | | hardcoded to "hostingInstance" |
| | type | String | | Hosting type name |
| | providerInstance | String | | Hosting provider instance name.  The provider could be of type  VM or an Internal Component. |
| | providerCompInstanceTask | String | yes | ComponentInstanceTask name - owner of the providerInstance |
| | consumerInstance | String | | Hosting consumer instance name |
| | consumerCompInstanceTask | String | yes | ComponentInstanceTask name - owner of the consumerInstance |
| HostingTypeTask | name | String | | Hosting name |
| | objType | String | | hardcoded to "hosting" |
| | provider | String | | Provider name |
| | providerCompTypeTask | String | yes | ComponentTypeTask name - owner of the Provider |
| | providedHostconfigName | String | yes | (Configuration) name |
| | providedHostdownloadCmd | String | yes | (Configuration) download command |
| | providedHostconfigureCmd | String | yes | (Configuration) configure command |
| | providedHostinstallCmd | String | yes | (Configuration) install command |
| | providedHoststartCmd | String | yes | (Configuration) start command |
| | providedHostStopCmd | String | yes | (Configuration) stop command |
| | providedHostuploadCmd | String | yes | (Configuration) upload command |
| | consumer | String | | Consumer name |
| | consumerCompTypeTask | String | yes | ComponentTypeTask name - owner of the Consumer |
| | requiredHostconfigName | String | yes | (Configuration) name |
| | requiredHostdownloadCmd | String | yes | (Configuration) download command |
| | requiredHostconfigureCmd | String | yes | (Configuration) configure command |
| | requiredHostinstallCmd | String | yes | (Configuration) install command |
| | requiredHoststartCmd | String | yes | (Configuration) start command |
| | requiredHostStopCmd | String | yes | (Configuration) stop command |
| | requiredHostuploadCmd | String | yes | (Configuration) upload command |